Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf____ 2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? No route__________ 0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 65________ 0

Does R2 show up as an OSPF neighbor on R1? Yes________ 2

Does R2 show up as an OSPF neighbor on R3? No ________ 2

What does this information tell you?

This indicates that passive interface configuration only affects the specified interface. R2's S0/0/0 interface actively sends Hello messages, so R2 maintains a neighbor relationship with R1; however, R2's S0/0/1 interface is configured as a passive interface and does not send Hello messages, so R2 and R3 cannot establish an OSPF neighbor relationship, R3 does not receive routing updates from R2, and therefore cannot learn routes from the 192.168.2.0/24 network. 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

router ospf 1

no passive-interface s0/0/1 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1________________

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65____

Is R2 listed as an OSPF neighbor to R3? Yes__________

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To more accurately calculate the cost of high-speed links and ensure that routing reflects actual link speed differences.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24 Cost 782 = Serial port cost from R1 to R3 (781) + Ethernet port cost of R3 (1)

192.168.23.0/30 Cost 845 = Serial port cost of R1 (781) + Serial port cost of neighboring router (64)

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new cost is 1562, because the bandwidth of all serial ports has been changed to 128Kb/s, so the cost of each serial port segment has increased from 64 to 781. R1 to 192.168.23.0/30 requires two serial port segments: 781 + 781 = 1562

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because R1 manually set the cost to 1565 on the S0/0/1 interface, which is higher than the cost of 1562 for the path through R2, according to the OSPF shortest path first algorithm, R1 chooses the path with the lower cost.

4

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   It can prevent router IDs from changing automatically due to interface status changes, avoid OSPF neighbor relationship resets, and facilitate network administrators in identifying and troubleshooting routers and ensuring that router IDs in the network are predictable.

   4

2. Why is the DR/BDR election process not a concern in this lab?

   Because all connections between routers in this experiment are point-to-point serial links. DR/BDR election only occurs in multi-access networks.

   4

3. Why would you want to set an OSPF interface to passive?

   The passive interface still advertises the network to other OSPF routers, while reducing unnecessary traffic, preventing the interface from sending Hello messages, reducing the CPU and memory load on the router, and preventing unauthorized routers from establishing OSPF neighbors.

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The Area design gives OSPF a hierarchical structure. By dividing the network into different areas, OSPF can support large-scale network deployments.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   EIGRP routes will be used because routers compare administrative distance rather than metric when selecting routes, and the default administrative distance for EIGRP (90) < the default administrative distance for OSPF (110) < the default administrative distance for RIP (120).